Pathophysiology of nutritional status and malassimilation syndrome

Content

- Regulation of appetite: orexins and anorexins
  - Central in hypothalamus
  - Peripheral: short-term and long-term effect

	central	peripheral
orexins	neuropeptide Y	ghrelin
anorexins	POMC producing peptide	Insulin, leptin

  - Pathophysiology of regulation of food intake and body mass: bulimia and others
- Basel metabolic rate
- Obesity: How we assess obesity (BMI, Tanita or Waist-to-hip ratio). Inflammation in obesity.
- Glycemic index
- Catabolic and anabolic state: BUN and creatinine, azotemia and uremia
- Starvation:
  - Short term
  - Long term: marasmus, kwashiorkor X Kwashiorkor-like syndrome, cachexia (e.g. anorexia Il-1 beta), sarcopenia, malabsorption (primary and secondary).

Appetite and satiety

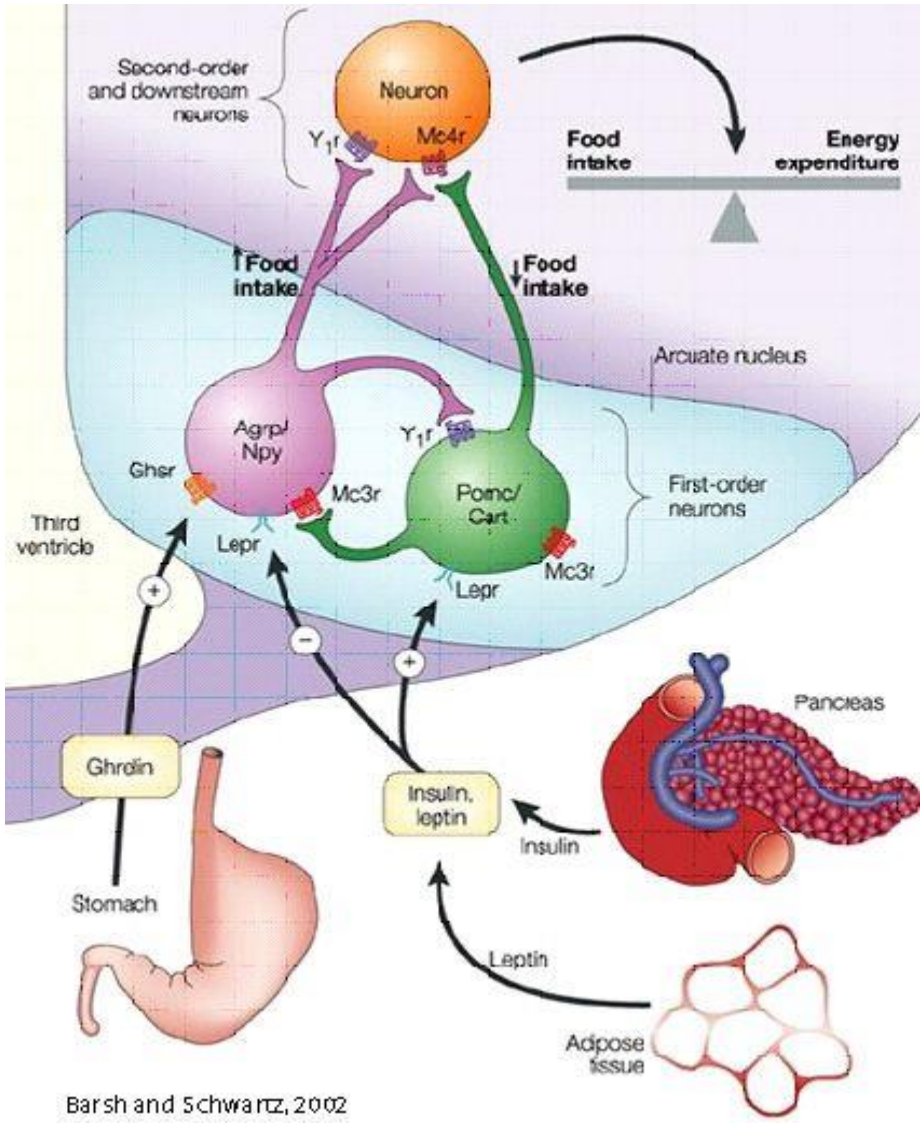
- Regulation of appetite and satiety occurs from the periphery acting on central controls including the brainstem, hypothalamus, and autonomic nervous system.

Central orexins and anorexins: Hypothalamus

- Orexins: which promotes appetite, and decreases metabolism (anabolic)
  - neuropeptide Y (NPY)
  - agouti-related protein (AGRP)

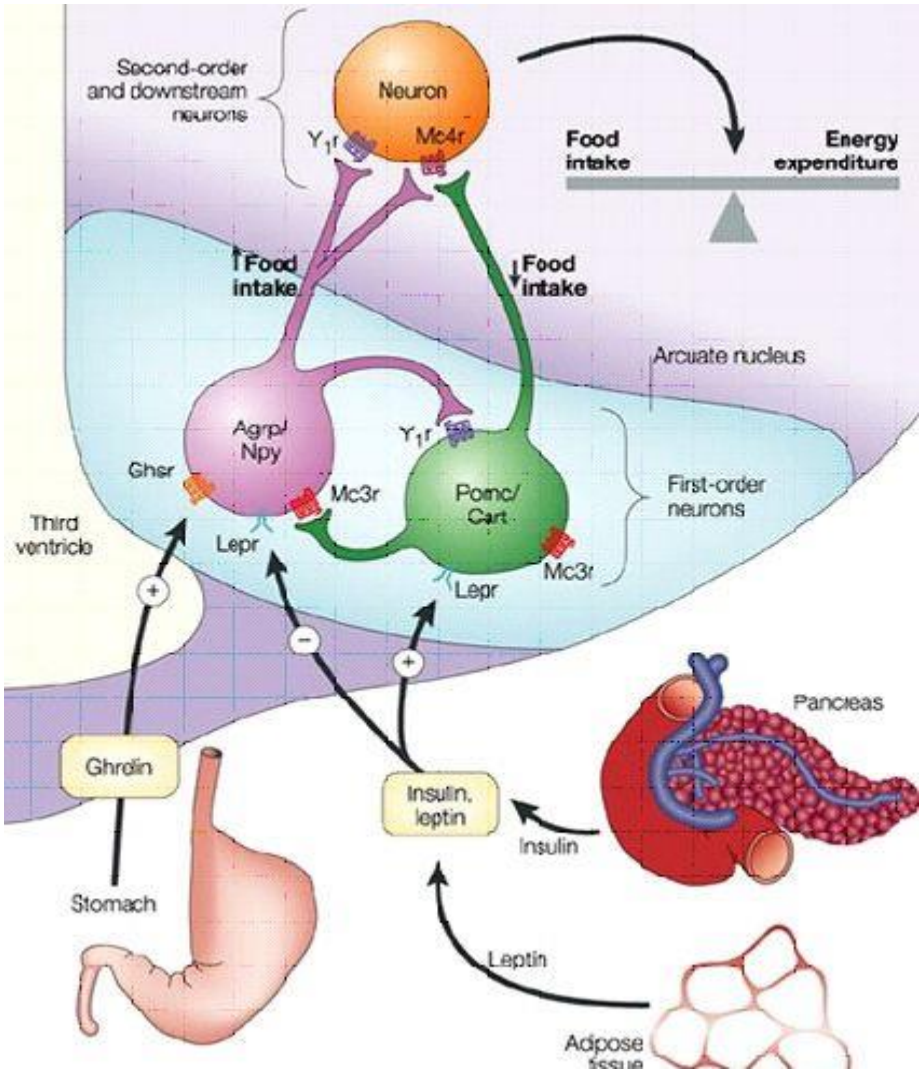
- Anorexins: which suppress appetite, and increase metabolism (catabolic).
  - cocaine-and-amphetamine–regulated transcript (CART)
  - pro-opiomelanocortin (POMC)–producing peptide

    - Proopiomelanocortin (POMC) is the pituitary precursor for
      - circulating melanocyte stimulating hormone (α-MSH)
      - adrenocorticotropin hormone (ACTH)
      - β-endorphin.



Peripheral orexin: Ghrelin from stomach

- Ghrelin is produced in the response to hunger (low sugar) in the stomach and stimulate appetite by activating neuropeptide Y-expressing neurons in the hypothalamus.
- Ghrelin induces hyperglycemia to keep homeostasis and antagonize lipolysis
- Ghrelin also stimulates release of growth hormone (GH) from anterior pituitary cells, release of gastric acid, gastrointestinal motility, and pancreatic functions.
- There is expected level in obesity (↓) and anorexia nervosa (↑).
  - ↑ in some obese individuals
  - ↓ associated with insulin resistance and hyperleptinemia in type 2 diabetes.



Orexin: Ghrelin from stomach



- **Prader–Willi syndrome** (PWS) is a genetic disorder caused by chromosome 15. It is associated with high ghrelin level and high appetite.

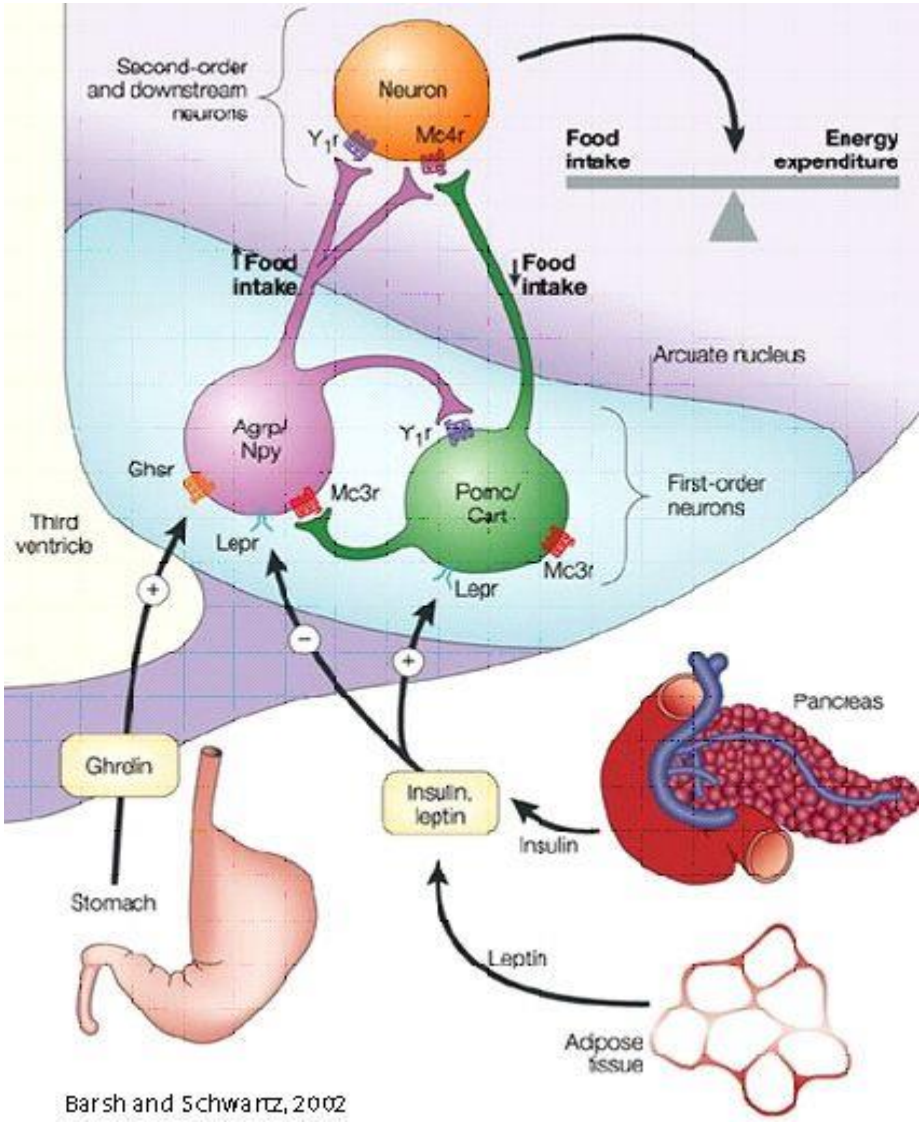
Anorexins

- Leptin and adiponectin from adipocytes
- glucagon-like peptide 1, intestinal peptide YY (PYY) and cholecystokinin (CCK) from the intestines
- insulin from pancreatic beta cells
- endocannabinoids from brain and peripheral nerves

Anorexins: leptin in long-term regulation

- High levels act via POMC producing peptide and suppress appetite and increase expenditure
- Low level induces neuropeptide Y and increase appetite
- In obesity, high level and leptin resistance (insulin resistance)
- In cachexia, low leptin does not induce neuropeptide Y

  - Another adipokines can contribute, e.g. adiponectin
    - Stimulation of glucose uptake, insulin sensitivity and lipid catabolism (β-oxidation)
    - Protection from endothelial dysfunction (prevention of atherosclerotic formation)
    - In obesity, decreased
    - In anorexia nervosa, increased but not from adipose tissue)



Incretin: Glucagon-like peptide 1

- It is produced and secreted by intestinal enteroendocrine L-cells and certain neurons in the brainstem upon food consumption.
- Incretin: decrease blood sugar levels in a glucose-dependent manner by enhancing the secretion of insulin.
- inhibits gastric emptying, acid secretion and motility collectively decreasing appetite.

Short term-anorexin cholecystokinin (CCK, pancreozymin)

- a hormone secreted especially by the duodenal mucosa in response to dietary products and nervous system (e.g. acetylcholine)
- improves digestion by
  - slowing down the emptying of food from the stomach
  - stimulating the production of bile in the liver as well as its release from the gall bladder (digestion of fats and proteins)

- a neuropeptide in the brain
- an anorexin: The mechanism for hunger suppression is thought to be a decrease in the rate of gastric emptying

Peptide YY

- released from cells in the ileum and colon in response to feeding.
- it slows the gastric emptying and inhibits gastric motility
- increases water and electrolyte absorption in the colon but suppress pancreatic secretion.
- obese people secrete less PYY than non-obese people

Pathophysiology of regulation of food intake and body mass.

- Anorexia nervosa
- Bulimia
- Binge eating
- Orthorexia refers to an unhealthy obsession with eating “pure” food. Food considered “pure” or “impure” can vary from person to person

Basal metabolic rate

- refers to the amount of calories that your body needs to perform basic, life-sustaining functions.

- It is the amount of energy expended at rest and neutral temperature. It is also necessary for the digestive system to be inactive, which requires at least a 12-hour fast. This number tells how much energy you need to sustain your vital organs like your brain, kidney, liver, heart, lungs, skin, muscles and your nervous system.

- It tells you how many calories your body would burn in a day if you do absolutely nothing and lie in bed.

Body mass index (BMI) and waist-to-hip ratios (waist circumference)

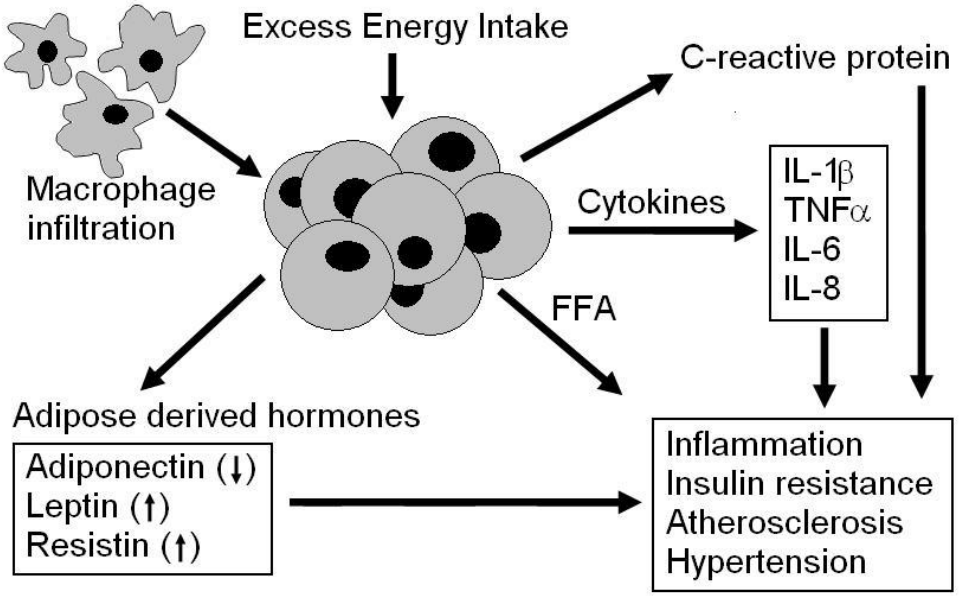
Obesity is defined as a body mass index (BMI) that exceeds 30 kg/m2 and generally develops when caloric intake exceeds caloric expenditure in genetically susceptible individuals.

Waist-to-hip ratios is useful in visceral and peripheral obesity

Obesity is a risk factor for hypertension, stroke, hyperlipidemia, gallstones, nonalcoholic steatohepatitis, gastroesophageal reflux, osteoarthritis, infectious disease, asthma, and obstructive sleep apnea.

Skinfold caliper





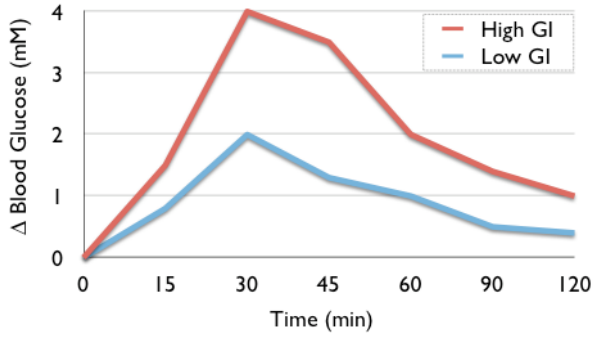
Tanita scale or InBODY

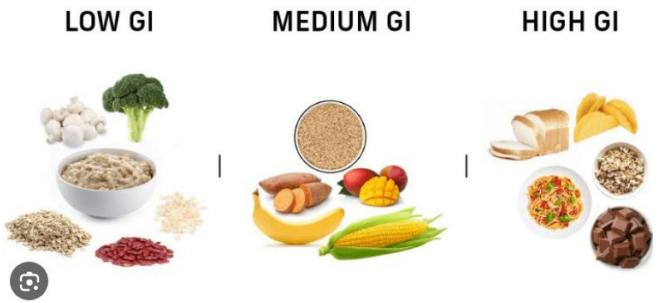
The bioimpedance scale is a diagnostic device that determine the percentage composition of the human body based on different tissue impedances. The method is based on supplying a weak alternating current to the organism. Thanks to the various ions, water behaves as a very good conductor. On the other hand, fat tissue has properties more like an insulator. We can therefore measure different impedances in different tissues and derive the relative or total values of the body composition from this.



Glycemic index

- The glycemic index is a value assigned to foods based on how quickly those foods cause increases in blood glucose levels. Foods low on the glycemic index (GI) scale tend to release glucose slowly and steadily. Foods with high glycemic index increase blood glucose rapidly. For instance, we would need to consume more beets than rice to cause the same spike in blood sugar levels.
- In particular, we measure the incremental area under the two-hour blood glucose response curve (AUC) following ingestion of a food with 50g of carbohydrate. The AUC of the test food is divided by the AUC of glucose (standard) and multiplied by 100the.
- Foods with a high GI are not necessarily unhealthy, and not all foods with a low GI are healthy. E.g. Fat and protein in a specific food can slow down carbohydrate absorption.





Fasting Glucose Level	Diagnosis
70-99 mg/dL (3.9 to 5.5 mmol/L)	Normal Fasting Blood Glucose
100-125 mg/dL (5.6 to 6.9 mmol/L)	Elevated Fasting Blood Glucose
>126 mg/dL (7.0 mmol/L) on two occasions	Diabetes

Nitrogen balance

- **Nitrogen Balance** = Nitrogen intake **-** Nitrogen loss (urinal urea and creatinine)
- Blood urea nitrogen (BUN) level and creatinine can be used in estimating nitrogen balance and can be used to study protein metabolism.

  – Positive nitrogen balance: growth, hypothyroidism, tissue repair, and pregnancy.
  – Negative nitrogen balance: burns, serious tissue injuries, fevers, hyperthyroidism, wasting diseases, during periods of fasting. A negative nitrogen balance can be used as part of a clinical evaluation of malnutrition.

Nitrogen: Creatinine and urea

- **Creatinine** is a breakdown product of creatine phosphate from muscle and protein metabolism. It is released at a constant rate by the body (depending on muscle mass).
    - Sarcopenia is associated with ↓ creatinine in blood and urine.
        - Creatine in meat is converted to creatinine on cooking, which is absorbed, causing increased serum creatinine.
    - ↓BUN levels indicate malnutrition and ↑ BUN indicates increased protein released from muscle tissue breakdown (autolysis) that occurs during starvation to provide an energy source.
- The BUN and creatinine is important for assessment of kidney function: ↑blood creatinine and ↑ BUN

Catabolic state: Uremia and Uremic syndrome

- Azotemia: less severe condition with increased levels of urea but is not yet so severe as to produce symptoms
- Uremia or Uremic syndrome: high levels of urea in the blood with symptoms.
  - The terminal clinical manifestation of kidney failure (also called renal failure)
  - Cause: Chronic renal insufficiency: Decreased glomerular filtration that leads to tubular compensation with altered phenotype of tubular cells and decreased number of glomeruli in chronic kidney

Starvation

- Sort-term starvation: glucose, hepatic glycogen (glycogenolysis for 12-24 hours) and after app. 8 hours gluconeogenesis from pyruvate, lactate, amino acids, glycerol portion of triglyceride (adrenaline, noradrenaline)

- Long-term starvation (3 days): fatty acids provide energy to cardiac and skeletal muscle cells and ketone bodies sustain brain tissue. Some glucose is still needed as fuel for brain tissue.

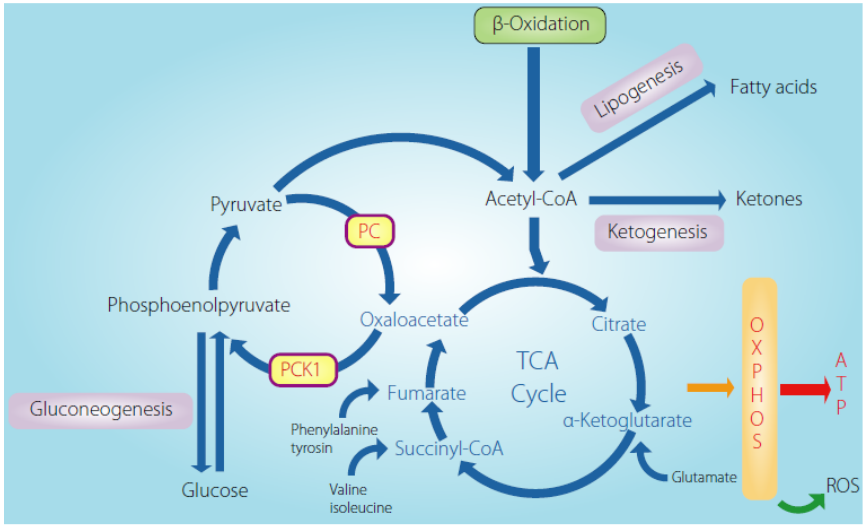


Figure 1 | Schematic of the hepatic tricarboxylic acid (TCA) cycle with the major anaplerotic and cataplerotic pathways associated with glucogenonesis, lipogenesis, and ketogeneis. An elevation

1. Simple starvation: Poor nutritional supplementation (e.g. poverty): **<u>marasmus</u>** without hepatomegaly (e.g. anorexia nervosa) and malnutrition such as **<u>kwashiorkor</u>** with hepatomegaly
2. Non-simple starvation:
    1. **<u>Kwashiorkor-like syndrome</u>**: stress (disease)
    2. chronic diseases leading to **<u>malabsorption</u>**
    3. chronic diseases leading to **<u>cachexia</u>** (e.g. anorexia Il-1 beta)
    4. aging or immobility leading to **<u>sarcopenia</u>**

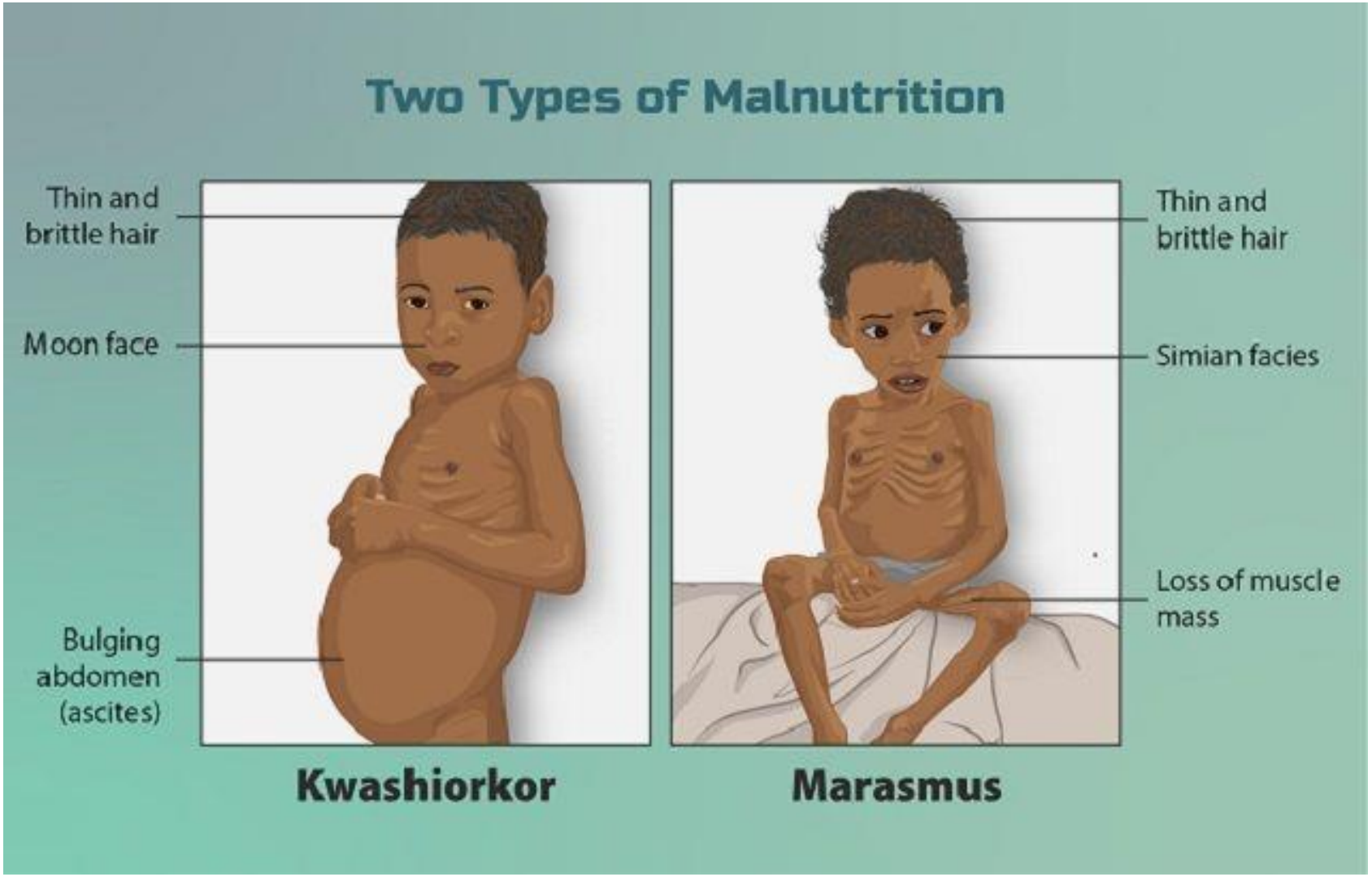
Starvation: hormones

- The short-term normal starvation: Initially, the level of insulin in circulation drops and the levels of glucagon, epinephrine, norepinephrine and GC rise: glycogenolysis, gluconeogenesis, lipolysis, and ketogenesis.

- Long-term normal starvation (simple): decreased circulating thyroid hormones, catecholamines and GC

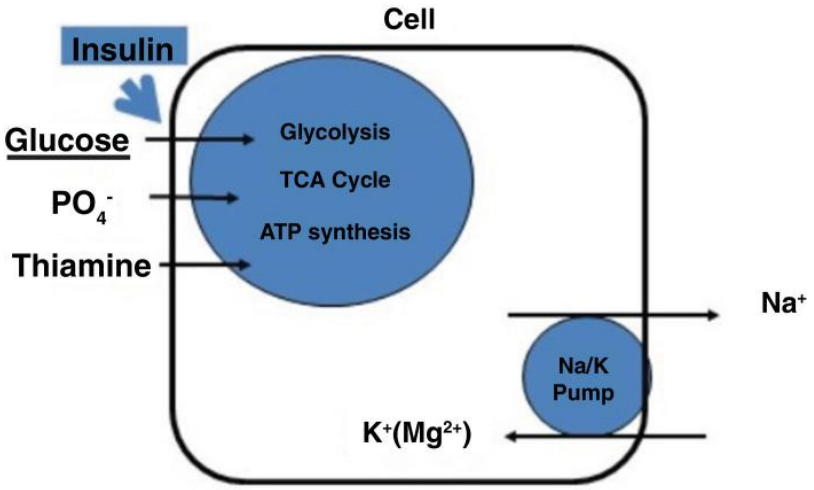
- Metabolic adaptation during stress: Catecholamines act transiently during the initial exposure to stress and at later stages would be accompanied by the secretion of glucocorticoids. GC promote hepatic gluconeogenesis and reduce glucose uptake and utilization in skeletal muscle and white adipose tissue. Chronic GC exposure results in hyperglycemia and insulin resistance.

Simple starvation: Marasmus X Kwashiorkor



Refeeding syndrome

- Hypophosphatemia (a biochemical hallmark), abnormal sodium and fluid balance; changes in glucose, protein, and fat metabolism; thiamine deficiency; hypokalemia; and hypomagnesemia.
- Oral feeding resulted in fatal diarrhea, vomiting, heart failure and neurological complication (coma).
  - Diarrhea may occur during the early stages of refeeding, due to atrophy of the intestinal mucosa and pancreatic impairment. The diarrhea generally resolves within the first few weeks of refeeding as the villous surface is reconstituted
- To prevent refeeding syndrome, initial fluid and electrolyte balance should be achieved prior to starting caloric intake. Once calories are started, the goal should be daily caloric requirements for age and weight divided into small frequent meals for the first one to two weeks.



Refeeding Syndrome

Refeeding syndrome occurs in severely malnourished individuals when parenteral or enteral nutritional therapy is initiated. During starvation, loss of body minerals causes the movement of phosphate, magnesium, and potassium ions out of the cells and into the plasma. When refeeding starts, an increase in insulin levels stimulates the movement of glucose and these ions back into the cells, and the plasma concentrations can decrease to dangerously low levels, causing hypophosphatemia, hypomagnesemia, hypokalemia, hyponatremia, hypocalcemia, and vitamin deficiency. Rapid expansion of the extracellular fluid volume also can occur with carbohydrate refeeding and may cause fluid overload. Hypophosphatemia contributes to alterations in red blood cell shape and function contributing to tissue hypoxia and increased respiratory drive. The consequence of these alterations includes life-threatening dysrhythmias, congestive heart failure, muscle weakness (including respiratory muscles), and death. Individuals at greatest risk are those with starvation from any cause including anorexia nervosa, chronic alcoholism, morbid obesity with massive weight loss, and prolonged fasting. Refeeding syndrome is prevented by slowly reinstituting feeding (about 20 kcal/kg/day for the first few days) and monitoring plasma levels of phosphate, potassium, magnesium, and calcium.

Stress starvation: Kwashiorkor-like syndrome: stress (disease): hormones

- Kwashiorkor-like syndrome is low circulating visceral proteins (albumin, retinol-binding protein, and transferrin) and a suppressed immune response caused by stress starvation that accompany advanced diseases, e.g. alcohol associated liver disease

Cachexia

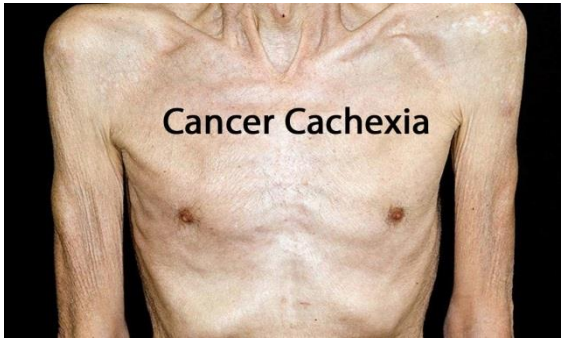
- A complex syndrome associated with an underlying illness causing ongoing muscle loss that is not entirely reversed with nutritional supplementation.

- Causes: end-stage cancer (upper gastrointestinal and pancreatic cancers), congestive heart failure, chronic obstructive pulmonary disease, chronic kidney disease and AIDS.

- Multifactorial pathophysiology:

1. The inflammation: TNF (which is also nicknamed 'cachexin' or 'cachectin') induces proteolysis and breakdown of myofibrillar proteins
2. Tumor-derived molecules (e.g. proteolysis-inducing factor) induce protein degradation
3. Alteration in feeding control loops in cachexia: high levels of leptin, a hormone secreted by adipocytes, block the release of neuropeptide Y leading to decreased appetite despite the high metabolic demand for nutrients

The eating disorders include anorexia nervosa, bulimia nervosa, and binge eating disorder. They are psychogenic disorders that can lead to malnutrition. Anorexia nervosa is the most serious and can be fatal.



Summary

	All nutrients	Proteins	Hormones
Poor diet	Marasmus	Kwashiorkor	↓GC, ↓thyroid hormones, ↓ catecholamine
Stress starvation in advanced diseases	Cachexia	• Visceral proteins: Kwashiorkor-like syndrome • Muscles: sarcopenia	↑GC
Normal acute stress			↑ epinephrine norepinephrine

Sarcopenia

- A type of skeletal muscle loss that occurs with aging and/or immobility (**loss of muscle mass and strength)**
  - the replacement of muscle fibers with fat and fibrosis
  - changes in muscle metabolism,
  - oxidative stress
  - degeneration of the <u>neuromuscular junction</u>







Handgrip

● APPENDIX: PHYSICAL STATUS ACCORDING TO THE TEST RESULT GIVEN BY THE DYNAMOMETER

AGE	MALE			FEMALE		
	Weak	Normal	Strong	Weak	Normal	Strong
10–11	‹12.6	12.6–22.4	›22.4	‹11.8	11.8–21.6	›21.6
12–13	‹19.4	19.4–31.2	›31.2	‹14.6	14.6–24.4	›24.4
14–15	‹28.5	28.5–44.3	›44.3	‹15.5	15.5–27.3	›27.3
16–17	‹32.6	32.6–52.4	›52.4	‹17.2	17.2–29.0	›29.0
18–19	‹35.7	35.7–55.5	›55.5	‹19.2	19.2–31.0	›31.0
20–24	‹36.8	36.8–56.6	›56.6	‹21.5	21.5–35.3	›35.3
25–29	‹37.7	37.7–57.5	›57.5	‹25.6	25.6–41.4	›41.4
30–34	‹36.0	36.0–55.8	›55.8	‹21.5	21.5–35.3	›35.3
35–39	‹35.8	35.8–55.6	›55.6	‹20.3	20.3–34.1	›34.1
40–44	‹35.5	35.5–55.3	›55.3	‹18.9	18.9–32.7	›32.7
45–49	‹34.7	34.7–54.5	›54.5	‹18.6	18.6–32.4	›32.4
50–54	‹32.9	32.9–50.7	›50.7	‹18.1	18.1–31.9	›31.9
55–59	‹30.7	30.7–48.5	›48.5	‹17.7	17.7–31.5	›31.5
60–64	‹30.2	30.2–48.0	›48.0	‹17.2	17.2–31.0	›31.0
65–69	‹28.2	28.2–44.0	›44.0	‹15.4	15.4–27.2	›27.2
70–99	‹21.3	21.3–35.1	›35.1	‹14.7	14.7–24.5	›24.5

Malassimilation syndrome

- Malassimilation is the decreased ability of the GI tract to incorporate nutrients into the body, either due to maldigestion or malabsorption.
    - Primary: Ciliac disease, Tropical sprue, Whipple's disease, Lactase deficiency or lactose intolerance
    - Secondary: Pancreatic insufficiency and Bile salt insufficiency

Lactase deficiency or lactose intolerance

- Congenital lactase deficiency (blacks)
- Acquired lactase deficiency:
- Risk factors: gluten-sensitive enteropathy, enteritis or bacterial overgrowth
- Undigested lactose leads to
  - Bacterial fermentation with gas production leads to bloating and flatulence
  - Osmotic diarrhea
  - Irritation and crampy pain

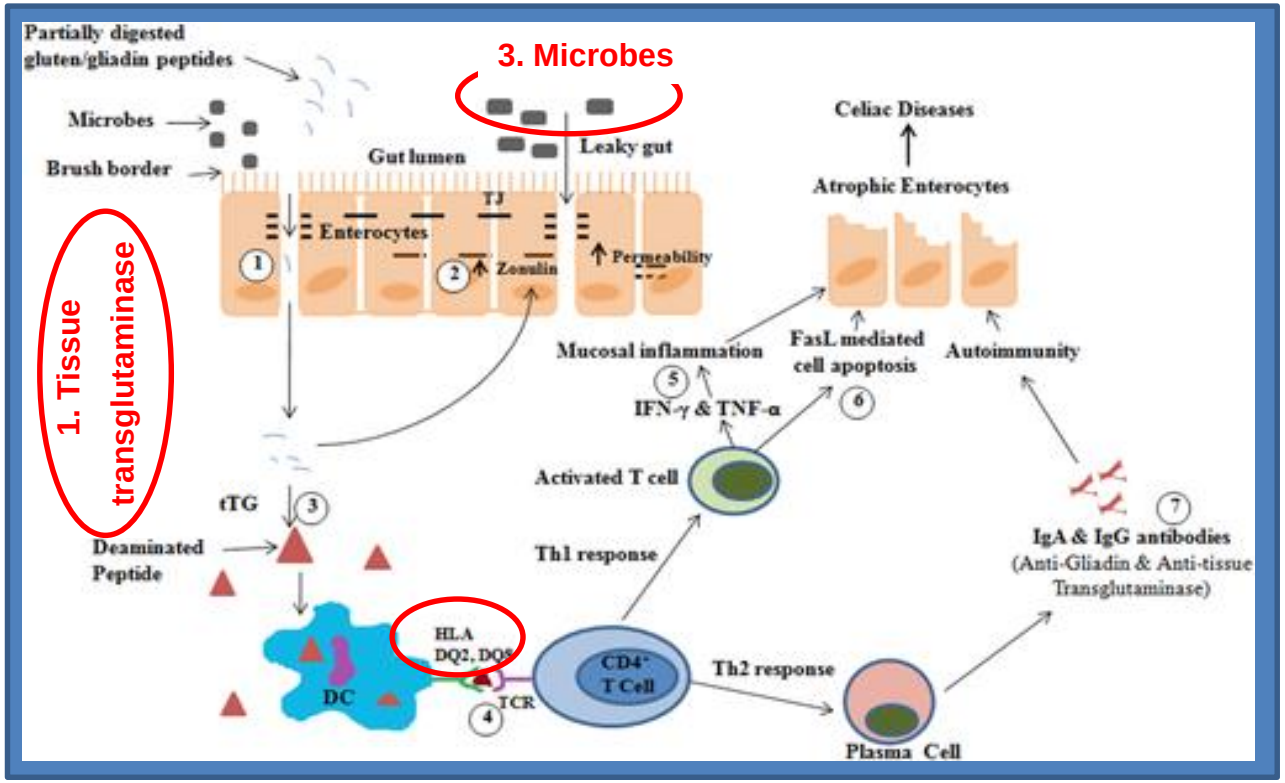
Ciliac disease, Celiac sprue, nontropical sprue, endemic sprue, gluten enteropathy

- Cause: abnormal reaction to gluten (gliadins and glutenins in corns or wheat) in genetically predisposed individuals (HLA-DQ)

- Diagnosis: anti-transglutaminase antibodies and biopsy (villous atrophy)

- Symptoms:
  - pale (anemia: iron, folic acid and vitamin B_{12} malabsorption)
  - calcium and vitamin D malabsorption (and compensatory secondary hyperparathyroidism) leading to osteopenia or osteoporosis
  - decreased vitamin K leading to petechia (hemostasis)
  - selenium malabsorption and copper and zinc deficiencies
  - loose (large volume), abnormally bad smelling and greasy stool (steatorrhoea) leading to weight loss or failure to gain weight
  - abdominal pain, cramping, bloating with abdominal distension (due to fermentative production of bowel gas)
  - later mouth ulcers, lactose intolerance and adenocarcinoma and lymphoma

Ciliac disease: pathophysiology



2. HLA-DQ2 allele or less HLA-DQ8 allele:
HLA-DQ is part of the MHC class II antigen-presenting receptor. HLA-DQ bind to gliadin peptides more tightly than other forms of the antigen-presenting receptor.

1. Anti-tissue transglutaminase antibodies (anti-tTG) : tTG catalize deamidation and transamidation of gluten. Transamidation forms the cross-linking of a glutamine residue from the gliadin peptide to a second protein with a lysine residue resulting in permanently covalently linked complex. This new epitopes is believed to trigger the primary immune response by which the autoantibodies against tTg develop

Tropical sprue

- A malabsorption disease commonly found in tropical regions, marked with abnormal flattening of the villi and inflammation of the lining of the small intestine.

- Signs and symptoms: diarrhoea, steatorrhoea, fever, malaise, weight loss, anorexia

- Causes:
  - not known
  - persistent bacterial, viral, amoebal, or parasitic infections.
  - folic acid deficiency
  - persistent small intestinal bacterial overgrowth

Whipple's disease

- A rare systemic infectious disease caused by the bacterium *Tropheryma whipplei*
- Signs and symptoms: diarrhea, abdominal pain, weight loss, and joint pains.

Pancreatic insufficiency

- Causes of pancreatic insufficiency include chronic pancreatitis, pancreatic carcinoma, pancreatic resection and cystic fibrosis.

- Although pancreatic insufficiency causes poor digestion of all nutrients, fat maldigestion is the chief problem. A large amount of fat in the stool (steatorrhea) and weight loss are the most common signs.

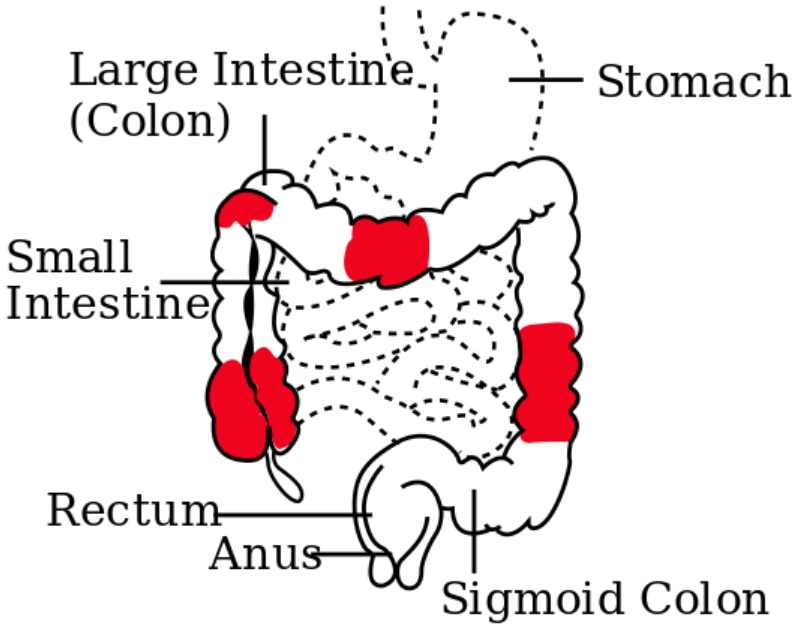
Bile salt insufficiency

- Causes of bile salt insufficiency include
  - advanced liver disease, which decreases production of bile salts
  - obstruction of the common bile duct, which decreases flow of bile into the duodenum;
  - intestinal stasis (lack of motility), which permits overgrowth of intestinal bacteria that deconjugate bile salts;
  - diseases of the ileum, which prevent the reabsorption and recycling of bile salts (enterohepatic circulation)
- Sign: poor intestinal absorption of fat (steatorrhea) and fat-soluble vitamins (A, D, E, and K).

Other courses of malassimilation: Crohn disease

- CD affects any part of the gastrointestinal tract from the mouth to the anus. The distal small intestine (malabsorption) and proximal large intestine are most commonly affected (patchy distribution in the colon or ileum).
- The typical chronic lesion is a skip granuloma having cobblestone projections of inflamed tissue surrounded by areas of ulceration. It starts in the intestinal submucosa and spreads across the intestinal wall to involve the mucosa and serosa in areas overlying lymphoid tissue. Complications include obstruction, fistulae, abscess formation, and chronic blood loss.
- Th1 immune response and IgG associated with severity
- Its causes are unknown: aggressive response to normal flora bacteria in genetically predisposed individuals (e.g. TLR)

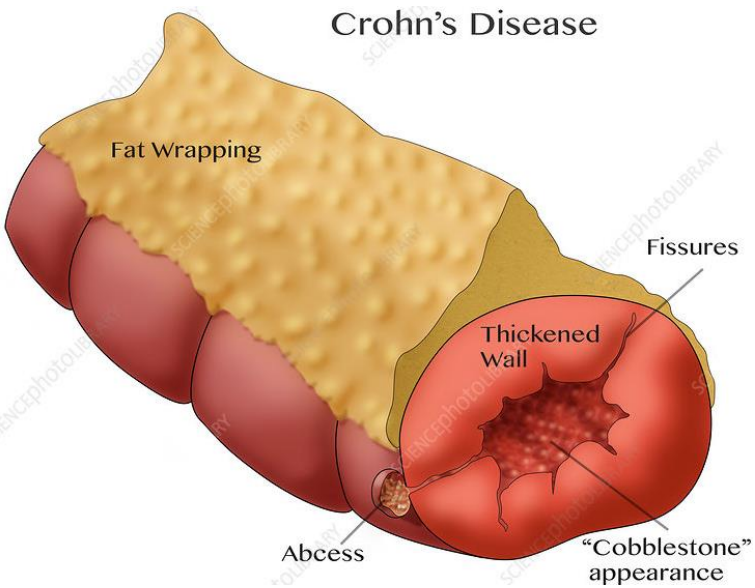
Crohn disease



Crohn's Disease

- The distal small intestine (malabsorption) and proximal large intestine are most commonly affected by patchy (local) distribution of lesions

Crohn disease: complications

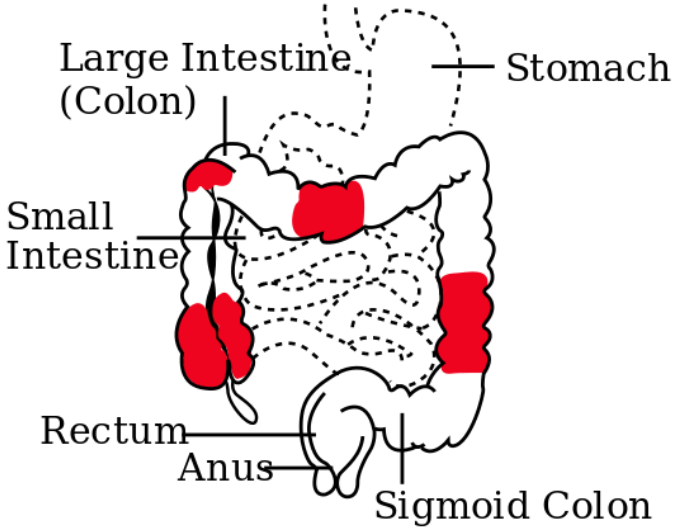




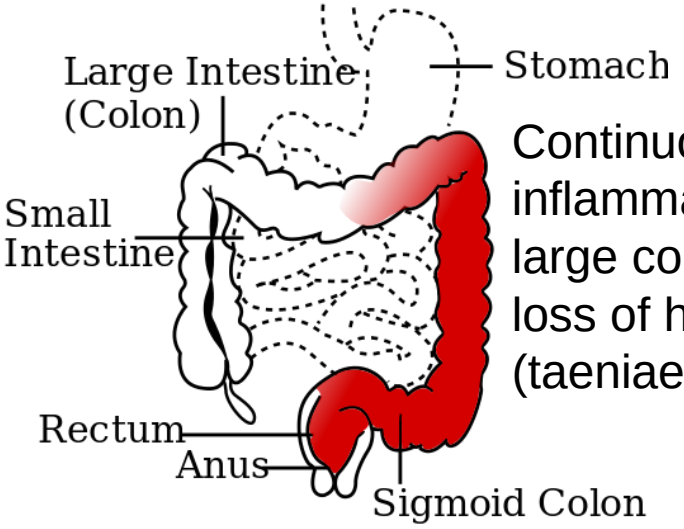


- It starts in the intestinal submucosa and spreads across the intestinal wall to involve the mucosa and serosa in areas overlying lymphoid tissue.
- Complications include obstruction, fistulae, abscess formation, and chronic blood loss.
  - The affected serosis leads to adhesions in which fistulas are formed.
  - The fibrosis leads to stenosis and colocky pain (subileus state)

Crohn disease x Ulcer colitis

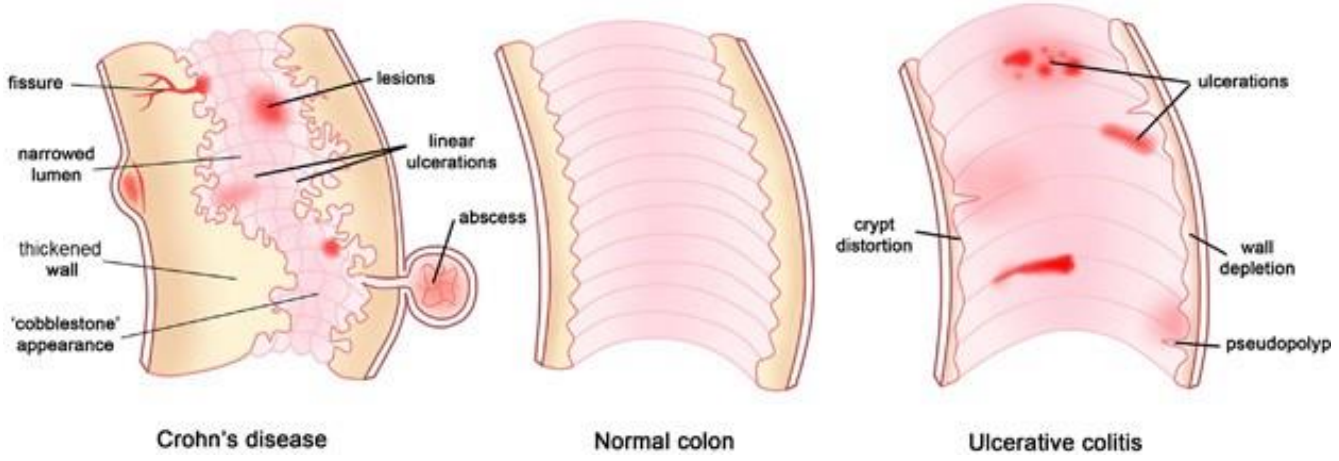


Crohn's Disease



Continuous inflammation in large colon and loss of haustra (taeniae relax)

Colitis ulcerosa



Other courses of malassimilation: Ulcerative colitis (UC)

- UC is a chronic inflammatory disease that causes ulceration of the colonic mucosa and extends proximally from the rectum into the colon: continuous lesions, are limited to the mucosa, and are not transmural. The mucous layer is thinner than normal bud edema and thickening of the muscular mucosae may narrow the lumen of the involved colon.
- In chronic disease, inflammatory polyps (pseudopolyps) develop in the colon from rapidly regenerating epithelium(also in CD). Mucosal destruction causes bloody diarrhea with passage of purulent mucus but severe hemorrhage is rare.
- T cells, plasma cells in the inflamed colon and colonic epithelial antibodies IgG in the circulation
- Its causes are unknown: bacterial, genetic, immunologic or dietary factors

References

Look up the particular theme in the following indexes

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